

A future for Rubi?

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1 Abstract

Although Albert Rich originally imagined that Rubi would be a system available to all mathematical users, the fact that it was developed using the pattern recognition capabilities of Mathematica has meant that without Mathematica, much of Rubi’s accumulated knowledge is not easy to access. Further, although much of the description and discussion of Rubi has highlighted the collection of over 5000 rules, the actual implementation requires in addition a very large number of auxiliary functions, and these also are written using Mathematica’s symbolic manipulation capabilities. Yet another challenge is the fact that Albert Rich emphasized in emails that he considered the *order* in which the rules were arranged was more important than the rules themselves.

It is a great misfortune that one of us (DJJ) had set aside time during his sabbatical leave to visit Albert in Hawaii and work with him on writing up his discoveries in this direction. Sadly, the sabbatical started in January, 4 months too late.

We have been re-implementing Rubi in Lisp and Reduce. This has required the translation of the rules from Mathematica syntax to Reduce syntax. This is going well, and already many integrals in the test suite are being evaluated. The eventual goal is a free, publicly accessible version of Rubi.

Keywords

Integration, Rule-based systems, Reduce System